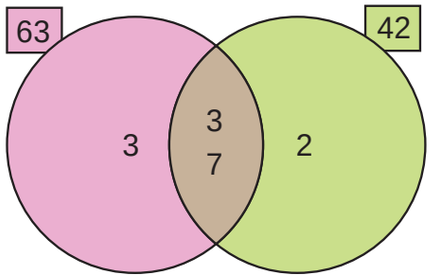
Illustration: Find the HCF of 63 and 42.

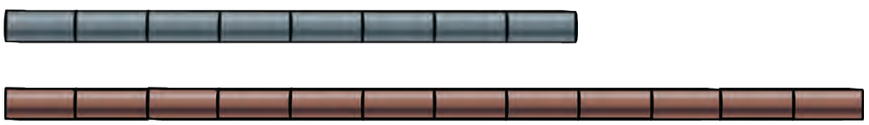
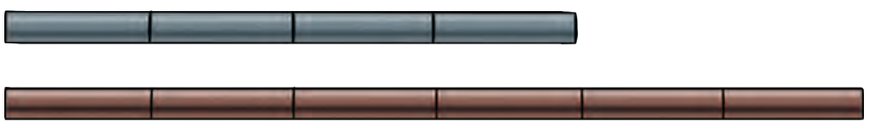
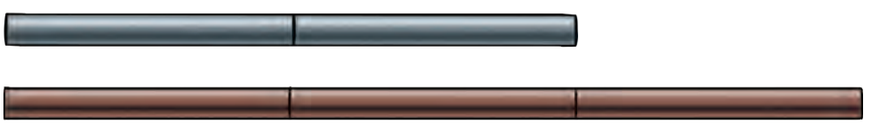


Solution: The prime factorisation of 63 is 3 x 3 x 7 and the prime factorisation of 42 is 2 x 3 x 7. We find that the common prime factors of 63 and 42 are 3 and 7 (see the diagram) and so the highest common factor is 3 x 7 =21. So, Pavithra can distribute equal number of athirasams (3 per family) and murukkus (2 per family) for a maximum of 21 families.

Situation 2:

Consider the rods of length 8 feet and 12 feet. We have to cut these rods into pieces of equal length. How many pieces can we get? What will be the length of the longest piece that is common for both the rods?

The rod of 8 feet can be divided into small rods of length 1 foot or 2 feet or 4 feet (These are factors of 8). The rod of 12 feet can be divided into small rods of length 1 foot or 2 feet or 3 feet or 4 feet or 6 feet (These are factors of 12). This is represented as follows:

	Number of pieces
First, it is possible to cut the 8 feet and 12 feet rods equally into 1 foot rods. 	8 12
Second, it is also possible to cut the 8 feet and 12 feet rods equally into 2 feet rods. 	4 6
Third, it is also possible to cut the 8 feet and 12 feet rods equally into 4 feet rods 	2 3

The length of the pieces that are common to both the rods (as given above) are of length 1 foot, 2 feet and 4 feet (i.e., common factors of 8 and 12).

Hence, the HCF of 8 and 12 is the length of the longest rod i.e., **4 feet** that can be cut equally from the rods of length 8 feet and 12 feet.

So, the **Highest Common Factor (HCF)** of two numbers is the largest factor that is common to both of them.The Highest Common Factor of the numbers *x* and *y* can be written as HCF (*x*,*y*).

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